13 **(a)** **(i)** Show that the cross-sectional area in Fig. C3.2 is $\pi x(2r-x)$. **[2]**

(ii) Hence show that the cross-sectional area is $\frac{\pi r^2}{h^2}(h^2-y^2)$, as given in line 37. **[2]**

(b) Verify that the formula $\frac{\pi r^2}{h^2}(h^2-y^2)$ for the cross-sectional area is also valid for

(i) Fig. C3.1, **[1]**

(ii) Fig. C3.3. **[1]**